

Dr Joshua Mark Pollin

Thermonuclear Supernovae: From 3D Explosion Models to Nebular Observables

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Education & Affiliations

Oregon State University , USA — Postdoctoral Researcher in Astrophysics	Jan 2026 – Present
Queen's University Belfast , UK — PhD Astrophysics	Oct 2022 – Sep 2025
- Thesis: <i>Exploring the Origins of Type Ia Supernovae with Monte Carlo Radiative Transfer</i> - Supervisors: Prof. S. Sim (QUB); Dr M. Nicholl (QUB)	
Queen's University Belfast , UK — Integrated MSci Physics (First Class Honours)	Sep 2018 – Jun 2022

Skills & Expertise

Simulation & Modelling: 3D time-dependent NLTE Monte Carlo radiative transfer (ARTIS); hydrodynamic core-collapse supernova modelling (PUSH); nucleosynthesis post-processing (CFNET).

HPC & Computing: C/C++, Python, Fortran, Bash, MPI, OpenMP; ~7M CPU hrs/yr personal allocation; Git, CI/CD.

Publications

First Author	Citations	Year
J. M. Pollin et al. <i>Updated 3D NLTE Nebular Diagnostics Favor D6 Models that Leave Behind a Surviving Companion for Normal Type Ia Supernovae</i> . RNAAS, Vol 10, No 7.	[-]	2026
J. M. Pollin et al. <i>Nebular Fingerprints of a Violent White Dwarf Merger: 3D NLTE Modelling of Type Ia Supernovae</i> . In review MNRAS.	[1]	2026
J. M. Pollin et al. <i>Multidimensional Nebular-Phase Calculations of Dynamically-Driven Double-Degenerate Double-Detonation Models for Type Ia Supernovae</i> . MNRAS 548, 1–32.	[14]	2025
J. M. Pollin et al. <i>On the Fate of the Secondary White Dwarf in Double-Degenerate Double-Detonation Type Ia Supernovae — II. 3D Synthetic Observables</i> . MNRAS 533, 3036–3052.	[28]	2024
Co-authored		
F. P. Callan et al. (incl. J. M. Pollin) <i>Multi-dimensional NLTE radiative transfer for a double detonation Type Ia explosion model in the photospheric phase</i> . MNRAS In Review.	[12]	2025
A. Holas et al. (incl. J. M. Pollin) <i>Observational Signatures of Thermonuclear Electron-Capture Supernovae</i> . A&A, Published June 2026.	[-]	2026
R. Pakmor et al. (incl. J. M. Pollin) <i>Violent Mergers Revisited: The Origin of the Fastest Stars in the Galaxy</i> . A&A 706, A239.	[8]	2026
A. Holas et al. (incl. J. M. Pollin) <i>The Asymmetry of White Dwarf Double Detonations and the Observed Scatter around the Phillips Relation</i> . A&A 698, A269.	[3]	2025
F. P. Callan et al. (incl. J. M. Pollin) <i>NLTE Spectral Modelling for a Carbon-Oxygen and Helium White Dwarf Merger as a Ca-Rich Transient Candidate</i> . A&A 702, A29.	[3]	2025
F. P. Callan et al. (incl. J. M. Pollin) <i>Exploring the Range of Impacts of Helium in the Spectra of Double Detonation Models for Type Ia Supernovae</i> . MNRAS 539, 1404–1413.	[12]	2025
S. Srivastav et al. (incl. J. M. Pollin) <i>Unprecedented Early Flux Excess in the Hybrid 02es-like Type Ia Supernova 2022ywc</i> . ApJ Lett. 956, L34. <i>Featured in Astrobites</i> .	[21]	2023
Software: ARTIS Collaboration (incl. J. M. Pollin), <i>artistools</i> / <i>artis</i> (Zenodo, 2024–2026).		

Grants & HPC Proposals

ESO VLT FORS2 — 4h service-mode time (P117) 2026
Co-Investigator on successful ESO VLT proposal P117 “Nebular Spectroscopy of Long-Duration Stripped-Envelope Supernovae” (PI: Thomas Moore). Awarded 4 hours of FORS2 service-mode observations.
Total HPC allocation across 4 national/international facilities: ~95M CPU hours; led science case on all proposals.

SuperMUC-NG (LRZ), Germany — 35M CPU hours	2025–2027
Co-authored proposal; led ~10% of science case. Delivered 1 first-author and 1 co-authored publication.	
DiRAC HPC Facility, UK — ~36M CPU hours	2024–2027
Co-authored proposal; led ~50% of science case. Delivered 1 first-author (9 citations), 1 first-author in preparation , and 2 co-authored publications.	
JUWELS (JSC/HITS), Germany — 21M CPU hours	2023–2025
Co-authored 2 successive proposals with HITS; led ~30% of science case. Delivered 2 co-authored publications.	
DiRAC HPC Facility, UK — 3M CPU hours	2023–2024
Sole science-case lead (100%). Delivered 1 first-author publication (21 citations).	

Scholarships & Awards

O’Sullivan Masterclass , Royal Irish Academy — One of ~20 selected from ~8,000 nationwide applicants; led by Nobel Laureate Prof. R. Genzel.	Apr 2025
Student Research Grant , Queen’s University Belfast — Led successful application to fund a summer research assistant; £2,000 awarded.	2024–2025
Emily S. Montgomery Travel Scholarship — £400 competitive travel award.	

Conference Talks & Organisation

[INVITED] Leiden, Netherlands	Jan 2027
[SEMINAR] Trinity College Dublin	Mar 2025
<i>Contributed & Organised</i>	
[ORGANISED] National Astronomy Meeting , Durham — proposed, organised & co-chaired session on explosive transients (~50–60 attendees)	Jul 2025
Rise Time Meeting, Purdue	Aug 2026
16th, 17th, 18th, 19th Würzburg Winter Workshop	Dec 2022–2025
RAS Specialist Discussion Meeting, London	Nov 2024
Oxford Transients Meeting	Jul 2024
European Workshop on White Dwarfs, Barcelona	Jul 2024
National Astronomy Meeting, Cardiff	Jul 2023

Teaching, Mentorship & Outreach

Aurora Outreach Video — Co-produced viral outreach video (~150,000 views)	May 2024
DiRAC Science Highlight — Late-time supernova simulations	Mar 2026
DiRAC Science Highlight — Photospheric-phase simulations	Mar 2024
Supervision of Research Projects , QUB — Microphysics in Type Ia simulations; 2 of 3 students progressed to PhD programmes	2024
Work Experience Mentoring , QUB — Mentored secondary school students; delivered seminars	Jul 2025
Graduate Teaching Assistant , QUB — Mathematics tutor; Teaching Fellowship Scheme	2023–2025
Support of Masters Students , QUB — Scientific guidance for 2 MSc projects	2023–2024
Girls in Maths & Physics , QUB — Annual talks and exoplanet workshops	2023–2025
NI Science Festival — Public lecture on using light to probe the Universe	Feb 2024
Irish Astronomical Society — Invited public talk on Type Ia supernovae	Jan 2024
Pathway Opportunity Programme , QUB — Computational workshops on cosmic expansion	2023–2024
Undergraduate Peer Mentor , QUB — Mentored incoming undergraduates	2019–2020

References

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